

13 Photosynthesis

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Photosystems in thylakoids absorb light, split water and make ATP and reduced NADP. The Calvin cycle in the stroma uses ATP and reduced NADP to fix CO₂ into triose phosphate; RuBP is regenerated. Light intensity, CO₂ and temperature may each limit rate, with changing limiting factor as conditions change.

Worked example: follow the reasoning

If CO₂ is limiting, brighter light may no longer increase photosynthesis because carbon fixation cannot use the extra ATP and reduced NADP quickly enough.

Quick check · multiple choice

A Where does Calvin cycle carbon fixation occur? [1]

- A Stroma
- B Thylakoid lumen
- C Mitochondrial matrix
- D Cytoplasm only

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A plant's oxygen production plateaus when light intensity doubles. State two possible limits and suggest one controlled test. [3]

Answers and explanation

Quick check: A. Stroma

Question 1 [3 marks]

CO₂ availability or enzyme temperature may limit [2]; vary CO₂ while holding light and temperature constant and observe whether oxygen production rises [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.